scRNA-seq analysis handbook

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Outline

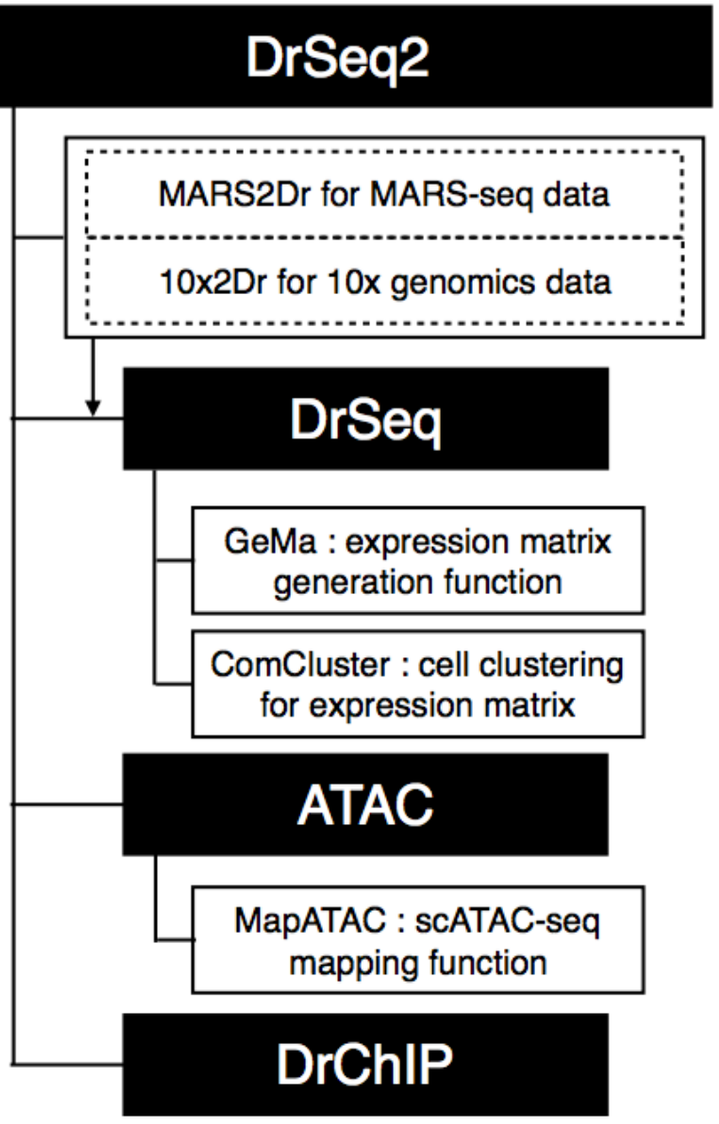
- scRNA-seq with reads barcoded
  - using Dr.seq2
- scRNA-seq without reads barcoded
  - see RNA-seq analysis

Dr.seq2 introduction

- Dr.seq2 is a QC and analysis pipeline for parallel single cell transcriptome and epigenome data. Dr.seq2 provides QC and analysis for three major data types: single cell transcriptome data (DrSeq part), Drop-ChIP data (DrChIP part) and scATAC-seq data (ATAC part).

- For single cell RNA-seq data, two additional step-by-step functions are included: 1. expression matrix generation for amounts of single cell RNA-seq datasets (GeMa step) and 2. cell clustering and analysis for the single cell expression matrix (comCluster step).

- For different single cell datasets, Dr.seq2 uses sequencing files in FASTQ format or SAM format as input with relevant command. For transcriptome data, DrSeq uses two FASTQ files as standard input (one file contains the transcript reads and the other one contains cell barcode information.



Dr.seq2 installation

STEP1.Install pipeline

1. Make sure you have Python (version 2.7) and R (version 2.14.1 or higher) in a Linux or MacOS environment.

2. Download Dr.seq2 from here.The github page of Dr.seq2 is https://github.com/ChengchenZhao/DrSeq2/



3. Install Dr.seq2 on your server/computer (please contact the administrator of that machine for assistance).

v2.1.1 (2017.05.04) Dr.seq2 new version release.

```
$ unzip Dr.seq2.1.2.zip
$ cd Dr.seq2.1.2    # find your Dr.seq2.1.2 folder and change working directory to it
```

for root user

```
$ sudo python setup.py install
```

If you are not a root user, Dr.seq2 can be installed at a specific location with write permission.

```
$ python setup.py install --prefix /home/DrSeq2    # here you can replace "/home/DrSeq2" with any location you want
$ export PATH=/home/DrSeq2/bin:$PATH    # setup PATH, so that system knows where to find executable files
$ export PYTHONPATH=/home/DrSeq2/lib/python2.7/site-packages:$PYTHONPATH    # setup PYTHONPATH, so that Dr.seq2 knows where to import modules
```

NOTE: To install Dr.seq2 on MacOSX, users need to download and install Xcode beforehand.

Dr.seq2 may take several minutes for installation.

Type:

```
$ DrSeq --help
```

If you see help manual, you have successfully installed Dr.seq2.

Prepare annotation and required softwares

STEP2.Prepare annotation and mapping software

1. Get gene annotation according to the species of your sample.
We provide convenient links for human (hg38) and mouse (mm10) at our webpage.You can download full annotation table from UCSC for other species (see Manual section)(Skip 2 and 3 if you already have bowtie2 and bowtie2 index)
hg38_refgenes.txt.zip on YunBaidu or on Google Drive(size:3.0M)
mm10_refgenes.txt.zip on YunBaidu or on Google Drive(size:2.6M)

2. Prepare mapping software(We use bowtie2(version 2.2.6) for users to get quick start. By default we use STAR(version >= 2.5.0) as aligner, see Manual:
We provide convenient links of executable bowtie2 for linux and MacOS user at our webpage. (otherwise you have to download full bowtie2 package and compile yourself, see Manual section):
Bowtie2 for linux-x86_64 on YunBaidu or on Google Drive
Bowtie2 for macos-x86_64 on YunBaidu or on Google Drive

For root user, you can just copy executable bowtie2 to any default PATH, (for example, /usr/local/bin)

```
$ unzip bowtie2-2.2.6-linux-x86_64.zip    # bowtie2-2.2.6-macos-x86_64.zip for macOS
$ cd bowtie2-2.2.6-linux-x86_64    # bowtie2-2.2.6-macos-x86_64 for macOS
$ sudo cp bowtie2* /usr/local/bin    # don't forget to type * mark here
```

If you are not root user, you can copy bowtie2 to the PATH (/home/drseq/bin) you setup in step1

```
$ unzip bowtie2–2.2.6-linux-x86_64.zip    # bowtie2-2.2.6-macos-x86_64.zip for macOS
$ cd bowtie2-2.2.6-linux-x86_64    # bowtie2-2.2.6-macos-x86_64 for macOS
$ cp bowtie2* /home/drseq/bin    # don't forget to type * mark here
```

type

```
$ echo $PATH
```

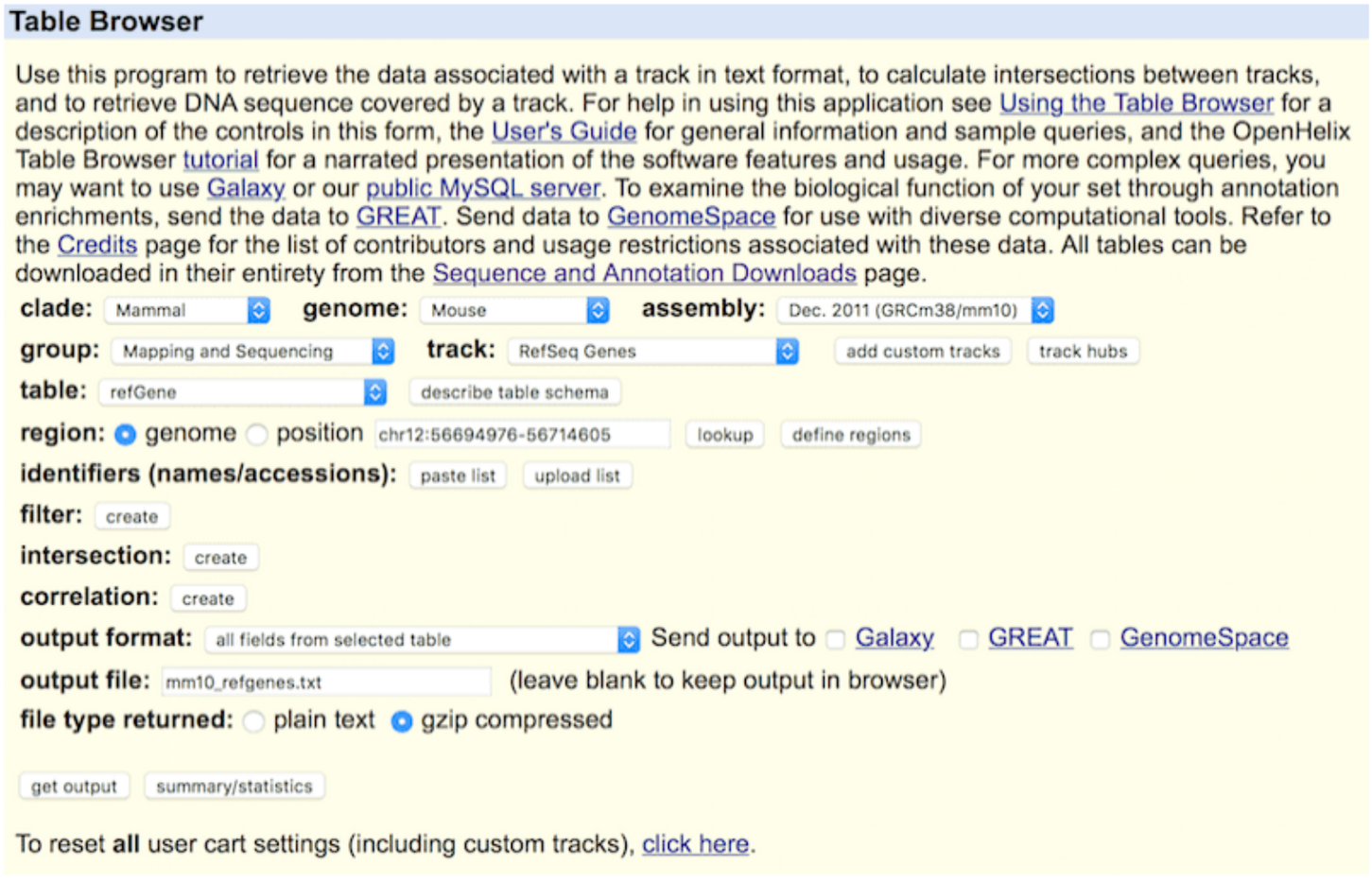
to check the list of your default PATH

3. Prepare bowtie2 index
We have provided a pre-built bowtie2 index at our webpage for a quick start. Downloading requires some time (you can build the bowtie2 index yourself if you are using a genome version other than hg19 or mm9, see the Manual section).
hg19 on YunBaidu or on Google Drive(size:3.7G)
mm9 on YunBaidu or on Google Drive(size:3.2G)

Prepare gene annotation file

Prepare gene annotation file

To use Dr.seq2 on transcriptome data, a gene annotation file with information regarding the location, name and composition of each gene is required. Dr.seq2 requires a gene annotation file with full information format (should be 16 coloumns) that can be downloaded from the UCSC genome browser http://genome.ucsc.edu/cgi-bin/hgTables. Below is an example for downloading the gene annotation file from the UCSC genome browser for the mm10 genome version.
We suggest using refseq genes for gene annotation, but you can also use annotations from other databases.



Run Dr.seq2 on single cell transcriptome data with reads barcoded in simple mode

Run Dr.seq2 on single cell transcriptome data

Standard mode and simple mode

The command in Dr.seq2 for running transcriptome data is DrSeq. DrSeq provides two modes. You can edit and design a parameter complex suitable for your data with the standard mode, but a configure file must be generated and edited before using DrSeq. Simple mode is designed for a convenient and quick start. With this mode, DrSeq can be run with a simple command line, but only major parameters can be edited with this mode.

Simple mode

The simple mode is described in the quick start section. In simple mode, only the input data, output name, mapping tool, annotation and mapping index must be specified with a command line, and DrSeq will generate QC and analysis reports with all default parameters.

All parameters used in the simple mode are also used in the standard mode and will be described below:

```
$ DrSeq simple [-h] -b BARCODE -r READS -n NAME[--cellbarcoderange CBR] [--umirange UMIR] [-g GA][--maptool
{STAR,bowtie2}] [--checkmem {0,1}][--mapindex MAPINDEX] [--thread P] [-f] [--clean][--select_cell_measure {1,2}][--
remove_low_dup_cell {0,1}]
```

Type

```
$ DrSeq simple -h
```

for detailed description of all major parameters of simple mode.

Run Dr.seq2 on single cell transcriptome data with reads barcoded in standard mode

Standard mode

In standard mode, you can edit all parameters and customize DrSeq for your data.
First, you should generate a configure file according to the template configure file (we will describe how to find and edit template configure files below) by typing:

```
$ DrSeq gen your_config_name.conf
```

Now you have generated a configure file (named your_config_name.conf in the example) according to the template configure file. If you open the configure file generated, you will find all changeable parameters listed with the pattern "parameter = value" (for example, barcode_file = SRR1853178_1.fastq) and the corresponding description in the bottom of each panel (starts with "#").

We use a Python package called "ConfigParser" to parse the configure file, which requires a specific pattern of the input configure file. Thus, users must take certain precautions when editing the configure file:
1. Do not edit the description lines (the lines starting with "#"); keep these lines that start with "#"
2. Keep the pattern "parameter = value" for parameter lines. Do not remove the value for any parameter line if you do not have a specific value for it. Keep the space next to the equal sign.

Now, you can edit your configure file and customize it for your Drop-ChIP data (If you do not have suitable initial parameters , you can run DrSeq in simple mode with only major parameters changed and edit other parameters according to the DrSeq output. Then, you can re-run DrSeq with updated parameters. Re-running DrSeq takes only 20% of the total time, see the FAQ section).

After you finish editing the configure the file, you can use standard mode to run DrSeq with your edited configure file input.

```
DrSeq run -c your_edit_config.conf -f --clean
```

-c your_edited_config.conf: name of your input configure file. The configure file should be generated using DrSeq gen mode or copied from another DrSeq run. The major parameters (parameters with "[required]" in the description line), including input file location, gene annotation file location, alignment software and mapping index, corresponding to the software should be specified by the user.
-f: force_overwrite: DrSeq will overwrite the output results if the output folder already exists when -f is added, otherwise DrSeq will exit.
--clean: DrSeq will remove intermediate result if --clean is added.

DIY template configure file to save your specific parameters

If it is too complicated to edit the configure file for the same parameters for each DrSeq run, the template configure file can be edited in the DrSeq package before DrSeq is installed to save changes to major parameters (for example, edit gene annotation file, alignment software, mapping index and other parameters) in the template configure file. If you have already installed DrSeq, the template configure file can be edited and installed again.

First, you should find the template configure file. It is located in the "lib/Config/" folder of the DrSeq package and named "DrSeq_template.conf" (do NOT change the file name of the template configure file).

After you find the template configure file, you can edit it to change the commonly used parameters. For example, if you change the parameter "gene_annotation = /data/mm10_refgenes.txt" and then re-install DrSeq, the configure file you generate next time will always have "gene_annotation = /data/mm10_refgenes.txt". Thus, you do not need to edit it again. Generally, we suggest editing gene_annotation, mapping_software_main and mapindex. You can also change cell_barcode_range if your barcode looks different from the published Drop-ChIP data (12 bp for cell barcode, 8 bp for UMI). Editing the template configure file is similar to editing the generated configure file.

Both standard mode and simple mode will use the edited the template configure file.

Cell clustering function

$comCluster -i ${expression_matrix} -o ${out_name} --cell col --seed 1 -d 1 -z 1.64

```
Usage: comCluster -i <input_expression_files> -o output_name -c clustering_method -d dimension_reduction_method -z
highvar_z -p select_pc_var --maxknum --custom_k --custom_d --rdnumber --cell  --seed

comCluster : clustering for multiple expression matrix
Options:
  --version             show program's version number and exit
  -h, --help            Show this help message and exit.
  -i INPUT_FILES, --input_files=INPUT_FILES
                        input files containing gene expression per
                        cell,multiple files should seperate by comma.This
                        parameter is required.
  -o OUTPUT_NAME, --output_name=OUTPUT_NAME
                        output name for summary files.This parameter is
                        required.
  -c CLUSTERING_METHOD, --clustering_method=CLUSTERING_METHOD
                        Method for cluster cells based on t-SNE output.This
                        parameter is required.Choose from 1(default) ,2 ,3 and
                        4.1(default) : k-means, use Gap statistics followed by
                        our 'first stable Gap' method to determine k;2 :
                        k-means, but use Gap statistics followed by
                        triditional 'Tibs2001SEmax'  method to determine k
                        (Tibshirani et al (2001));3 : k-means with custom
                        determined k, k value is defined in following
                        parameter 'custom_k', this option is designed for
                        users who know the number of subgroup of the drop-seq
                        sample;4 : (make sure your R enviroment has library
                        'fpc' installed) Use dbscan as clustering method, the
                        (eps) parameter is defined in following parameter
                        'custom_d'
-d DIMENSION_REDUCTION_METHOD, --dimension_reduction_method=DIMENSION_REDUCTION_METHOD
                        Method of dimension reduction,1 strands for 't-SNE',2
                        strands for 'SIMLR',3 strands for 'PCA'.This parameter
                        is required.Default is 1.
-z HIGHVAR_Z, --highvar_z=HIGHVAR_Z
                        Z-score cutoff for highly variable genes selection.
-p SELECT_PC_VAR, --select_pc_var=SELECT_PC_VAR
                        Select topN PC until they explain 50% (default is 0.5,
                        eg. 0.3 for 30% variance) of total variance.
--maxknum=MAXKNUM       Maximum k number for gap statistics, only take effect
                        when clustering_method = 1 or 2.
--custom_k=CUSTOM_K     Only take effect when clustering_method = 3, cells
                        will be clustered to N group based on t-SNE result
                        according to user determined k.
--custom_d=CUSTOM_D     Only take effect when clustering_method = 4, refer to
                        the parameter 'eps'(Reachability distance, see Ester
                        et al. (1996)) of dbscan. By default we set it to 2
                        according to the orginal Drop-seq paper, but it varies
                        a lot between differen datasets.
--rdnumber=RDNUMBER     Set random number.
--cell=CELL             Whether a column or a row strand for a cell.'col' for
                        column and 'row' for row.This parameter is required.
--seed=SEED             Set seed to keep your result reproducible
```